

HealthGov

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```
# Load Libraries
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.4.3
```

```
library(ggplot2)

# Read only column D from Excel
gender_data <- read_excel(
  "C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
  col_names = TRUE,
  range = cell_cols("D")
)

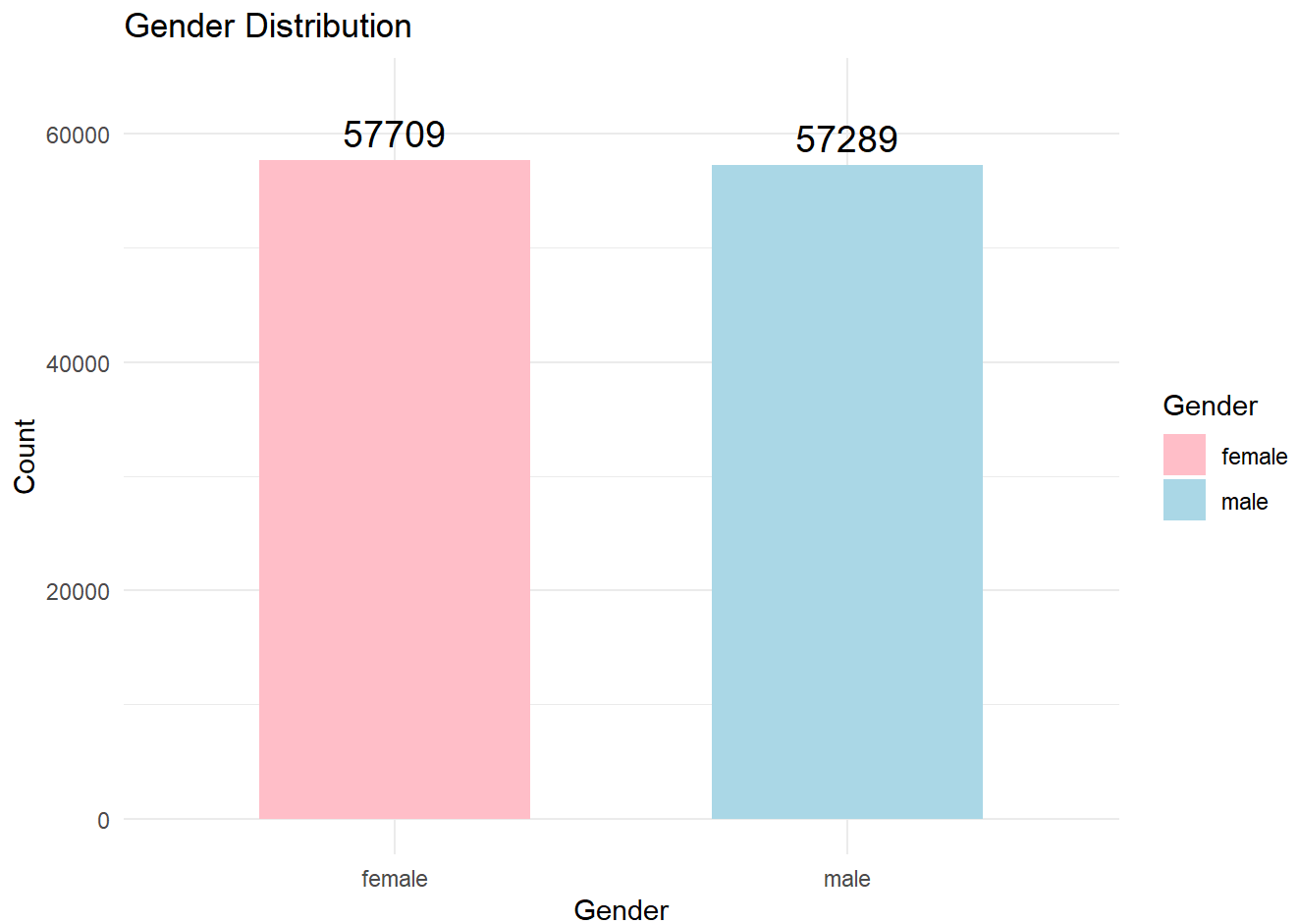
# Clean whitespace
gender_data[] <- lapply(gender_data, function(x) if (is.character(x)) trimws(x) else x)

# Count male/female directly
gender_counts <- as.data.frame(table(gender_data[[1]]))
colnames(gender_counts) <- c("Gender", "Count")

# Print to check
print(gender_counts)
```

```
##   Gender Count
## 1 female 57709
## 2   male 57289
```

```
# Plot
ggplot(gender_counts, aes(x = Gender, y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 0.6) +
  geom_text(aes(label = Count), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c("female" = "pink", "male" = "lightblue")) +
  labs(
    title = "Gender Distribution",
    x = "Gender",
    y = "Count"
  ) +
  coord_cartesian(ylim = c(0, max(gender_counts$Count) * 1.1)) + # safer than ylim()
  theme_minimal()
```



```
# Load Libraries
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)

# Read Gender and Ethnicity
data <- read_excel(
  "C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
  col_names = TRUE,
  range = cell_cols(c("D", "G")) # Replace with your actual Ethnicity column
)

# Clean whitespace
data[] <- lapply(data, function(x) if (is.character(x)) trimws(x) else x)

print(names(data))
```

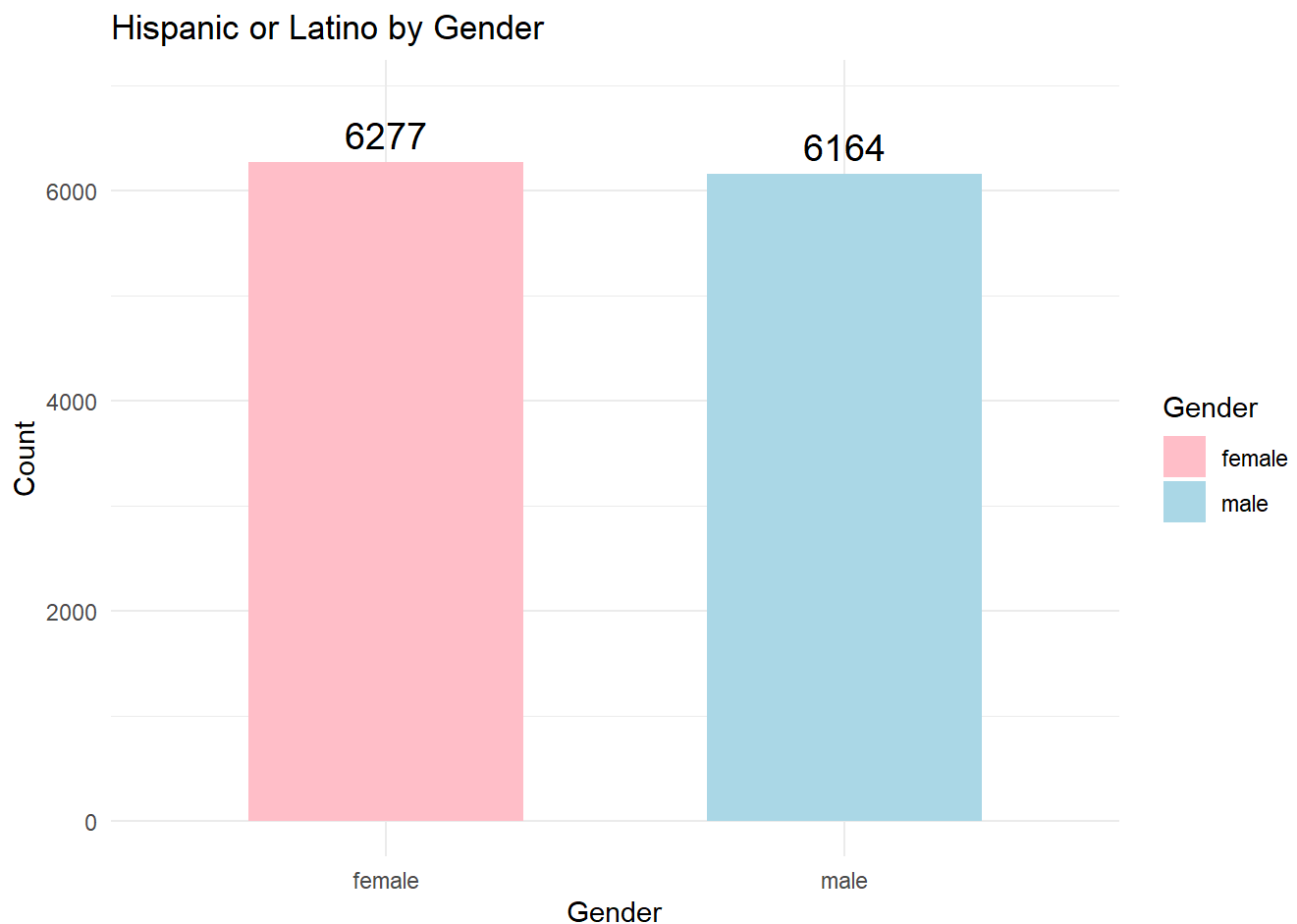
```
## [1] "Gender"      "BirthDate"   "MaritalStatus" "Ethnicity"
```

```
# Filter and count
ethnicity_counts <- data %>%
  filter(Ethnicity == "Hispanic or Latino") %>%
  group_by(Gender) %>%
  summarise(Count = n()) %>%
  ungroup()

print(ethnicity_counts)
```

```
## # A tibble: 2 × 2
##   Gender Count
##   <chr>   <int>
## 1 female  6277
## 2 male   6164
```

```
# Plot
ggplot(ethnicity_counts, aes(x = Gender, y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 0.6) +
  geom_text(aes(label = Count), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c("female" = "pink", "male" = "lightblue")) +
  labs(
    title = "Hispanic or Latino by Gender",
    x = "Gender",
    y = "Count"
  ) +
  coord_cartesian(ylim = c(0, max(ethnicity_counts$Count) * 1.1)) +
  theme_minimal()
```



```
# Load Libraries
library(readxl)
library(ggplot2)

# Read Marital Status column (F)
marital_data <- read_excel(
  "C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
  col_names = TRUE,
  range = cell_cols("F")
)

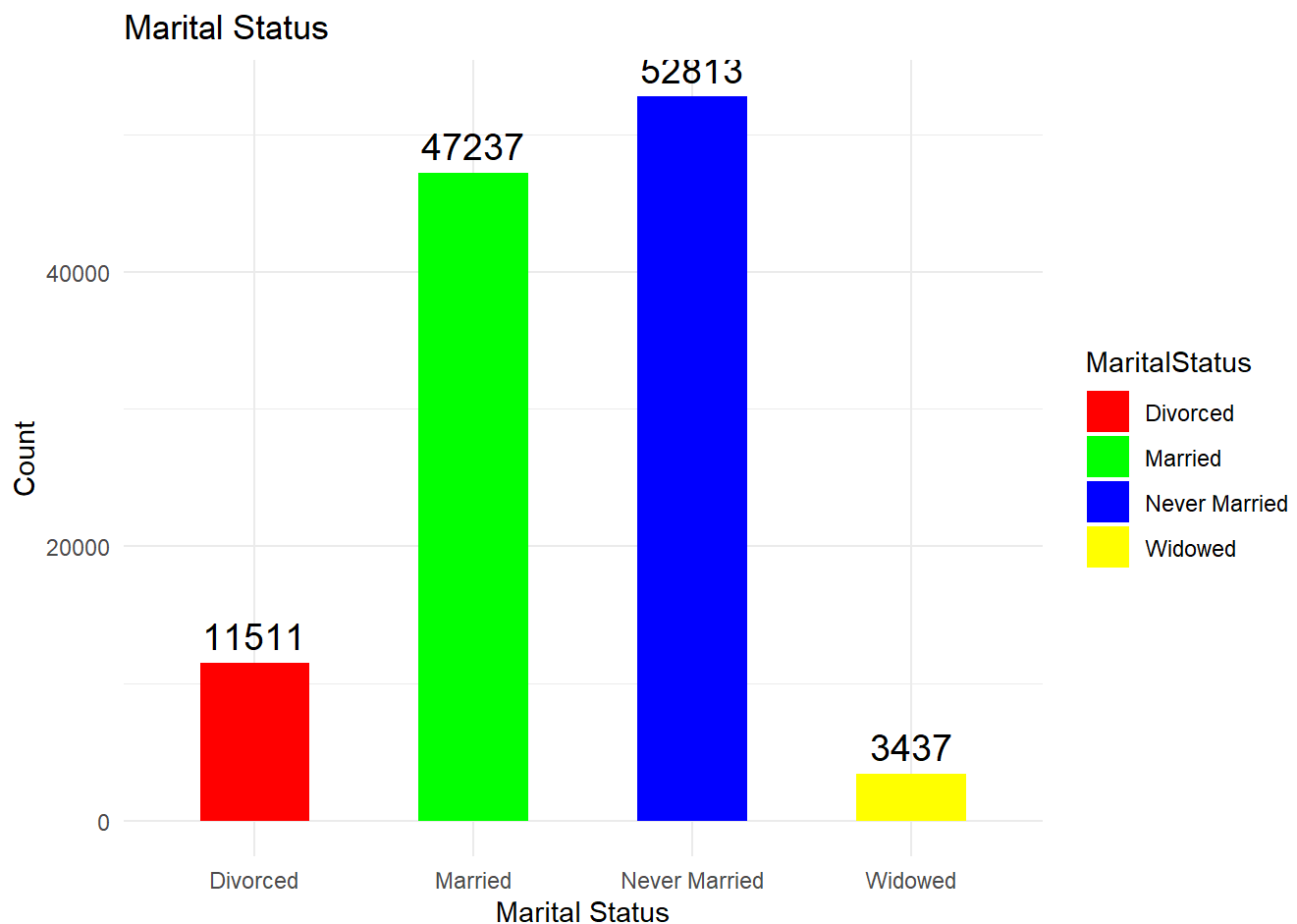
# Clean whitespace
marital_data[] <- lapply(marital_data, function(x) if (is.character(x)) trimws(x) else x)

# Count Marital Status
marital_counts <- as.data.frame(table(marital_data[[1]]))
colnames(marital_counts) <- c("MaritalStatus", "Count")

print(marital_counts)
```

```
## MaritalStatus Count
## 1 Divorced 11511
## 2 Married 47237
## 3 Never Married 52813
## 4 Widowed 3437
```

```
# Plot
ggplot(marital_counts, aes(x = MaritalStatus, y = Count, fill = MaritalStatus)) +
  geom_bar(stat = "identity", width = 0.5) +
  geom_text(aes(label = Count), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c(
    "Married" = "green",
    "Never Married" = "blue",
    "Widowed" = "yellow",
    "Divorced" = "red"
  )) +
  labs(
    title = "Marital Status",
    x = "Marital Status",
    y = "Count"
  ) +
  theme_minimal()
```



```
## Load Libraries
library(readxl)
library(dplyr)
library(stringr)

# Read the Excel file
data <- read_excel("C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
  col_names = TRUE)

# 1 Trim whitespace for all character columns
data[] <- lapply(data, function(x) if (is.character(x)) str_trim(x) else x)

# 2 Standardize Gender
data$Gender <- str_to_title(data$Gender)
# e.g., 'male', 'MALE' -> 'Male'
data$Gender[data$Gender %in% c("M", "m")] <- "Male"
data$Gender[data$Gender %in% c("F", "f")] <- "Female"

# 3 Standardize MaritalStatus (title case)
data$MaritalStatus <- str_to_title(data$MaritalStatus)

# 4 Calculate Age (from BirthDate)
data$BirthDate <- as.Date(data$BirthDate)
data$Age <- as.numeric(format(Sys.Date(), "%Y")) - as.numeric(format(data$BirthDate, "%Y"))

# 5 Group Age
below_25 <- sum(data$Age < 26, na.rm = TRUE)
between_26_50 <- sum(data$Age >= 26 & data$Age <= 50, na.rm = TRUE)
above_50 <- sum(data$Age > 50, na.rm = TRUE)

# 6 Count Gender
male <- sum(data$Gender == "Male", na.rm = TRUE)
female <- sum(data$Gender == "Female", na.rm = TRUE)

# 7 Summarize DisabilityAdjustedLifeYears
dalys_low <- sum(data$DisabilityAdjustedLifeYears < 5, na.rm = TRUE)
dalys_mid <- sum(data$DisabilityAdjustedLifeYears >= 5 & data$DisabilityAdjustedLifeYears < 10,
  na.rm = TRUE)
dalys_high <- sum(data$DisabilityAdjustedLifeYears >= 10, na.rm = TRUE)

# 8 Summarize QualityAdjustedLifeYears
qalys_low <- sum(data$QualityAdjustedLifeYears < 5, na.rm = TRUE)
qalys_mid <- sum(data$QualityAdjustedLifeYears >= 5 & data$QualityAdjustedLifeYears < 10, na.rm
  = TRUE)
qalys_high <- sum(data$QualityAdjustedLifeYears >= 10, na.rm = TRUE)

# 9 Total Conditions & Medications
total_conditions <- sum(data$ConditionCount, na.rm = TRUE)
total_medications <- sum(data$MedicationCount, na.rm = TRUE)
```

```
#  Print summary  
cat("Gender Counts:\n")
```

```
## Gender Counts:
```

```
cat("Male:", male, "\n")
```

```
## Male: 57289
```

```
cat("Female:", female, "\n\n")
```

```
## Female: 57709
```

```
cat("Age Groups:\n")
```

```
## Age Groups:
```

```
cat("Below 25:", below_25, "\n")
```

```
## Below 25: 31874
```

```
cat("26-50:", between_26_50, "\n")
```

```
## 26-50: 34023
```

```
cat("Above 05:", above_50, "\n\n")
```

```
## Above 05: 49101
```

```
cat("DisabilityAdjustedLifeYears:\n")
```

```
## DisabilityAdjustedLifeYears:
```

```
cat("<5:", dalys_low, "\n")
```

```
## <5: 100169
```

```
cat("5-10:", dalys_mid, "\n")
```

```
## 5-10: 5797
```

```
cat(">10:", dalys_high, "\n\n")
```

```
## >10: 9032
```

```
cat("QualityAdjustedLifeYears:\n")
```

```
## QualityAdjustedLifeYears:
```

```
cat("<5:", qalys_low, "\n")
```

```
## <5: 6993
```

```
cat("5-10:", qalys_mid, "\n")
```

```
## 5-10: 6245
```

```
cat(">10:", qalys_high, "\n\n")
```

```
## >10: 101760
```

```
cat("Total Conditions:", total_conditions, "\n")
```

```
## Total Conditions: 4216239
```

```
cat("Total Medications:", total_medications, "\n")
```

```
## Total Medications: 3812852
```



```
# Load libraries
library(readxl)
library(dplyr)
library(ggplot2)
library(stringr)

# Read your data
data <- read_excel("C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
                  col_names = TRUE)

# Clean up
data[] <- lapply(data, function(x) if (is.character(x)) str_trim(x) else x)
data$MaritalStatus <- str_to_title(data$MaritalStatus)
data$DisabilityAdjustedLifeYears <- as.numeric(data$DisabilityAdjustedLifeYears)

# Create a new column: Above5
data$DALYs_Group <- ifelse(data$DisabilityAdjustedLifeYears > 5, "Above 5", "5 or Below")

# Filter for the marital statuses you want
status_list <- c("Divorced", "Married", "Never Married", "Widowed")

# Loop through each status and plot pie chart
for (status in status_list) {

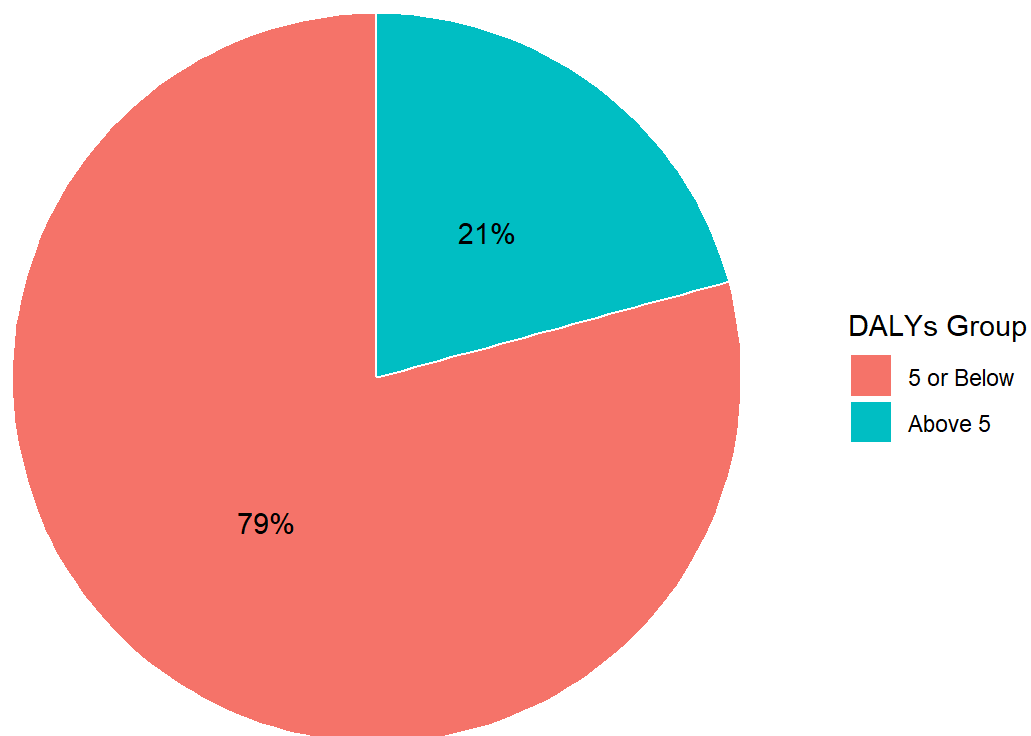
  pie_data <- data %>%
    filter(MaritalStatus == status) %>%
    group_by(DALYs_Group) %>%
    summarise(Count = n())

  # Make sure percentages add up
  pie_data$Fraction <- pie_data$Count / sum(pie_data$Count)

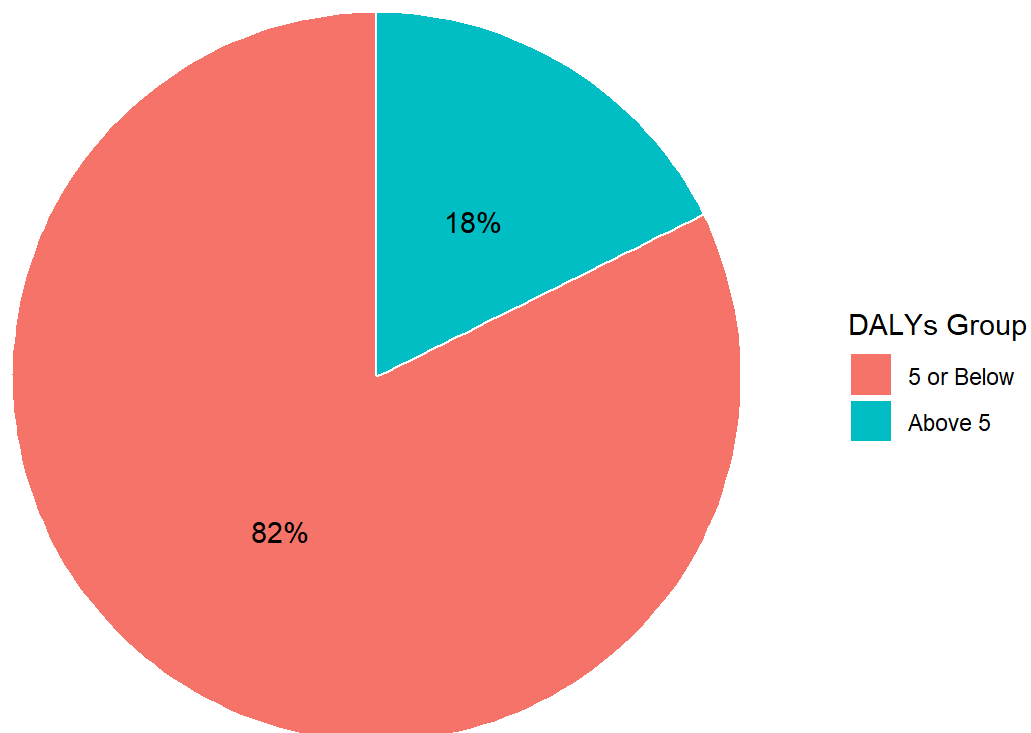
  # Plot pie chart
  p <- ggplot(pie_data, aes(x = "", y = Fraction, fill = DALYs_Group)) +
    geom_bar(stat = "identity", width = 1, color = "white") +
    coord_polar(theta = "y") +
    labs(
      title = paste("DALYs Above 5 vs 5 or Below -", status),
      fill = "DALYs Group"
    ) +
    geom_text(aes(label = scales::percent(Fraction)),
              position = position_stack(vjust = 0.5)) +
    theme_void()

  print(p)
}
```

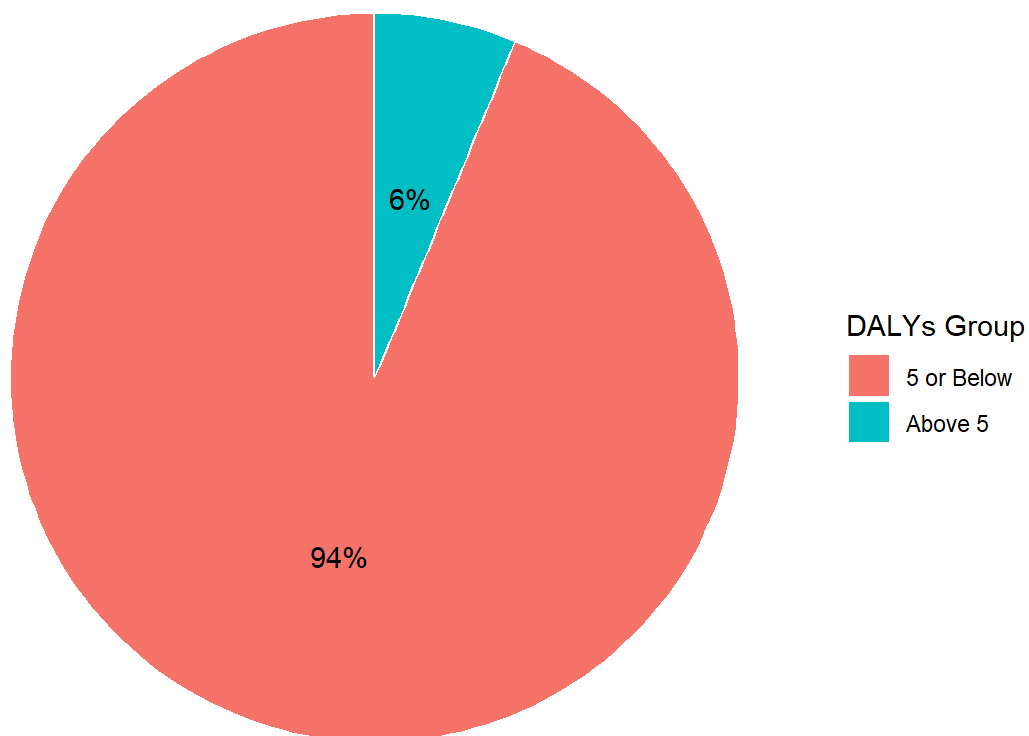
DALYs Above 5 vs 5 or Below - Divorced



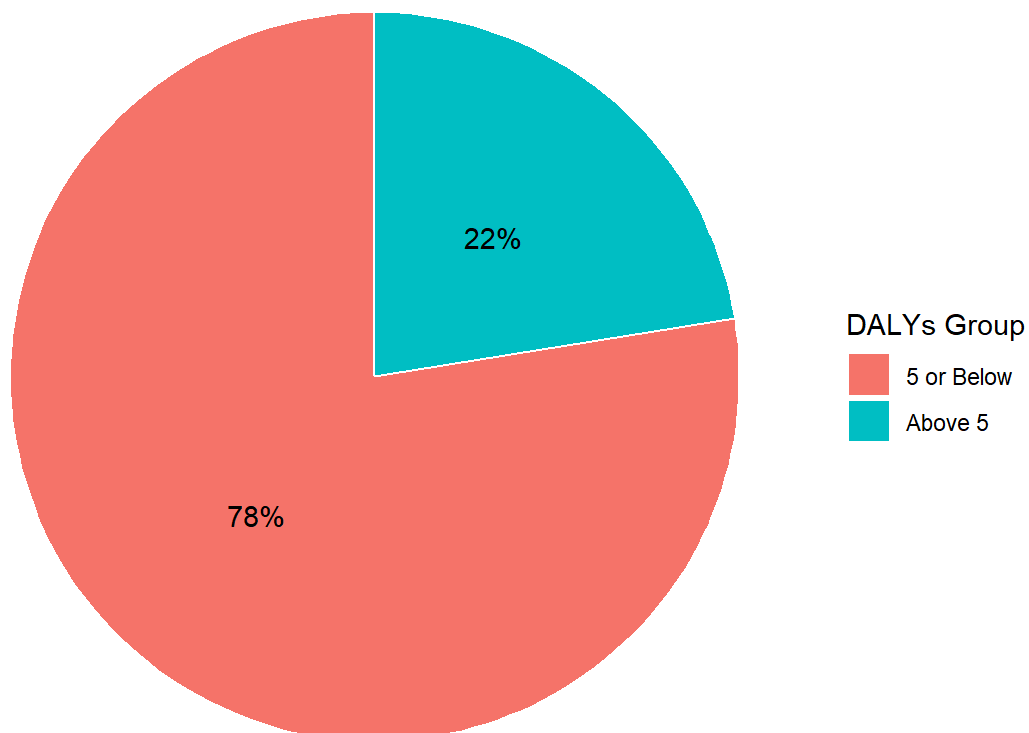
DALYs Above 5 vs 5 or Below - Married



DALYs Above 5 vs 5 or Below - Never Married



DALYs Above 5 vs 5 or Below - Widowed



```

# Load libraries
# Load libraries
library(readxl)
library(dplyr)
library(ggplot2)
library(stringr)

# Read data
data <- read_excel("C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx",
                  col_names = TRUE)

# Clean whitespace & convert DALYs
data[] <- lapply(data, function(x) if (is.character(x)) str_trim(x) else x)
data$DisabilityAdjustedLifeYears <- as.numeric(data$DisabilityAdjustedLifeYears)

# Create DALYs group
data$DALYs_Group <- ifelse(
  data$DisabilityAdjustedLifeYears >= 0 & data$DisabilityAdjustedLifeYears <= 1,
  "0-1",
  "Other"
)

# Summarize counts
dalys_counts <- data %>%
  group_by(DALYs_Group) %>%
  summarise(Count = n())

# Add the new "4x Others" bar
other_count <- dalys_counts$Count[dalys_counts$DALYs_Group == "Other"]

dalys_counts <- rbind(
  dalys_counts,
  data.frame(DALYs_Group = "Ideal", Count = 4 * other_count)
)

print(dalys_counts)

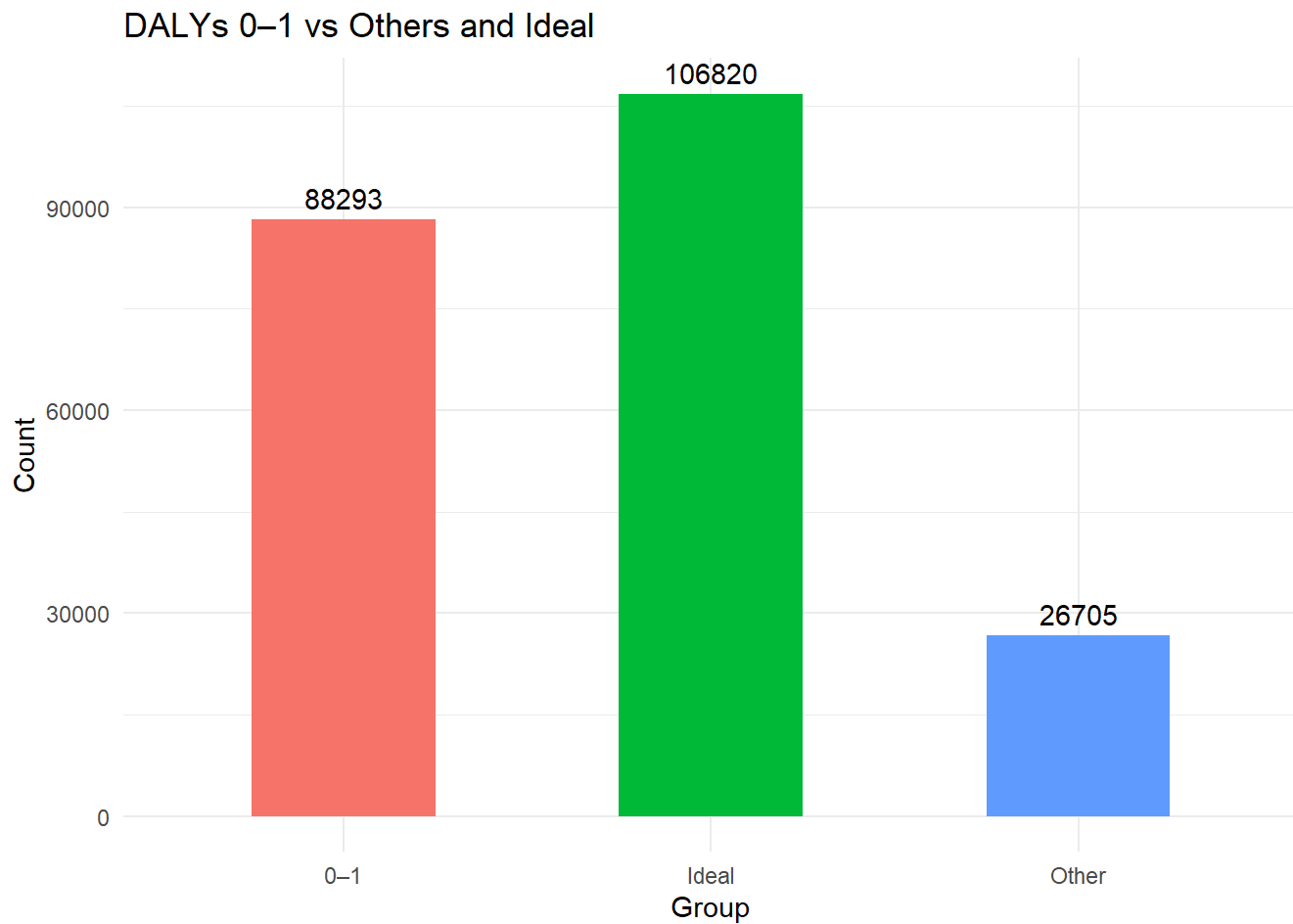
```

```

## # A tibble: 3 × 2
##   DALYs_Group Count
##   <chr>      <dbl>
## 1 0-1        88293
## 2 Other     26705
## 3 Ideal    106820

```

```
# Plot
ggplot(dalys_counts, aes(x = DALYs_Group, y = Count, fill = DALYs_Group)) +
  geom_bar(stat = "identity", width = 0.5) +
  labs(
    title = "DALYs 0-1 vs Others and Ideal",
    x = "Group",
    y = "Count"
  ) +
  geom_text(aes(label = Count), vjust = -0.5) +
  theme_minimal() +
  theme(legend.position = "none")
```



```
# Load Libraries
library(readxl)
library(ggplot2)
library(dplyr)
library(stringr)

# Read data
data <- read_excel("C:/Users/ojasb/OneDrive/Desktop/Prakat/HealthGov/synthea_output.xlsx")

# Clean
data[] <- lapply(data, function(x) if (is.character(x)) str_trim(x) else x)
data$ConditionCount <- as.numeric(data$ConditionCount)
data$DisabilityAdjustedLifeYears <- as.numeric(data$DisabilityAdjustedLifeYears)

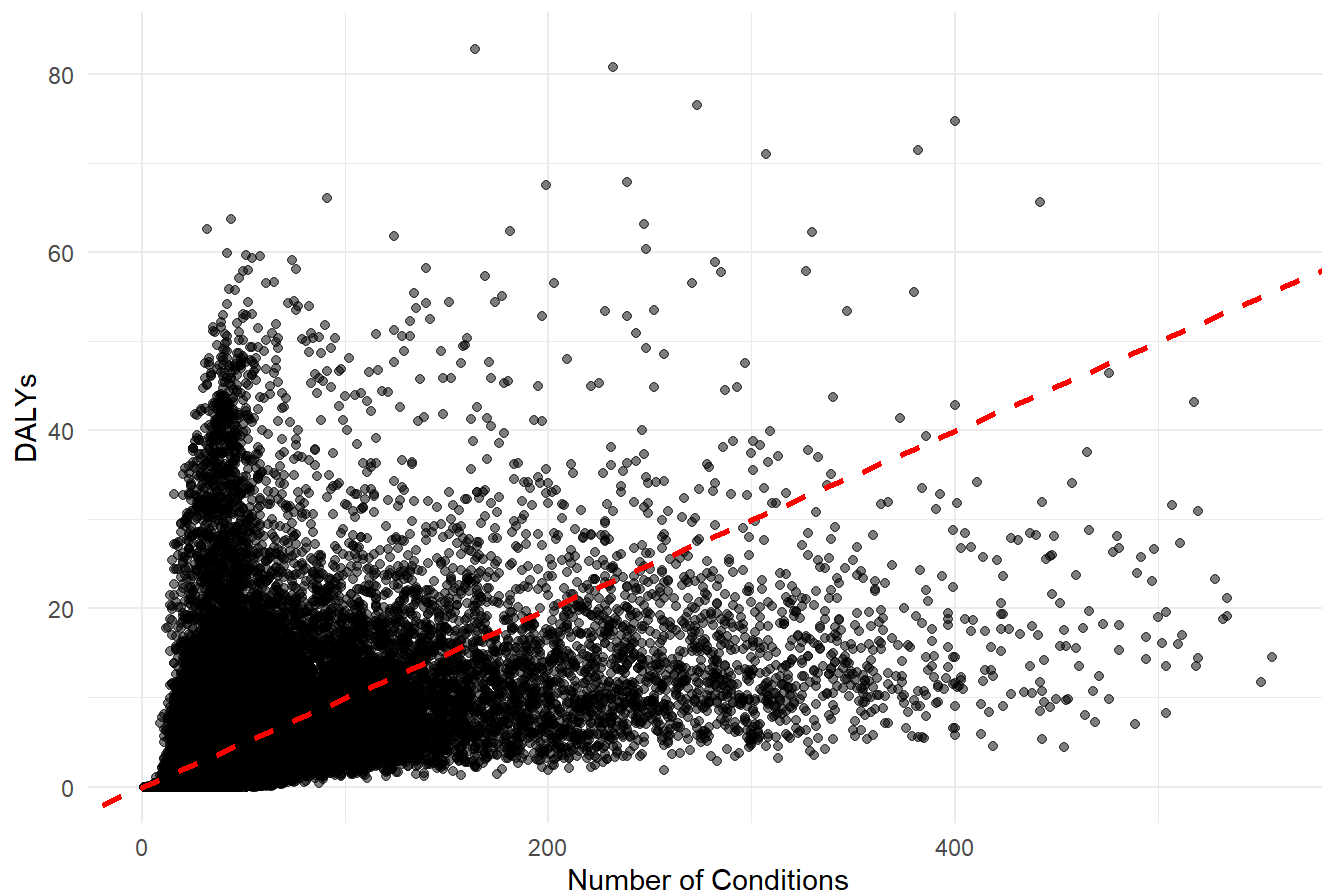
# Scatter plot
ggplot(data, aes(x = ConditionCount, y = DisabilityAdjustedLifeYears)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "loess") +
  geom_abline(intercept = 0, slope = 0.1, color = "red", linetype = "dashed", size = 1) +
  labs(
    title = "Condition Count vs. Disability Adjusted Life Years",
    x = "Number of Conditions",
    y = "DALYs"
  ) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Failed to fit group -1.
## Caused by error in `predLoess()`:
## ! workspace required (19837931284) is too large probably because of setting 'se = TRUE'.
```

Condition Count vs. Disability Adjusted Life Years



```
# Make sure numeric
data$MedicationCount <- as.numeric(data$MedicationCount)

# Scatter plot
ggplot(data, aes(x = ConditionCount, y = MedicationCount)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "loess") +
  labs(
    title = "Condition Count vs. Medication Count",
    x = "Number of Conditions",
    y = "Number of Medications"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Failed to fit group -1.
## Caused by error in `predLoess()`:
## ! workspace required (19837931284) is too large probably because of setting 'se = TRUE'.
```

Condition Count vs. Medication Count

